MILITARY SPECIFICATION

TRITIUM (HYDROGEN 3) SOLID FORM, SEALED SELF LUMINOUS SOURCES APPLICABLE FOR MILITARY MATERIEL: GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers minimum requirements for sealed (encapsulated) sources of tritium activated luminous material, including identification, marking and handling procedures. The sealed tritium activated luminous material shall hereinafter be referred to as "sealed material".

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

(Copies of standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATIONS

Code of Federal Regulations (Atomic Energy) Title 10

(Applications for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification of Manufacturer.- Manufacturers or producers shall not submit any materiel containing tritium activated luminous material until a specified license has been obtained from the Atomic Energy Commission as prescribed in the Code of Federal Regulations (Atomic Energy) Title 10 - Part 30.

3.2 Identification, License and Disposition.- When the sealed material is used with military materiel or equipment, the identification, license number and disposition for disposal shall be stamped or otherwise marked by decal, molding or other approved means to the materiel or equipment in the immediate area of the sealed luminous materiel. In addition, the month and year of manufacture shall appear together with the radiation caution symbol of any color, as specified by the contracting officer.

3.3 Material.- The luminous material shall consist of a smooth homogenous mixture consisting of essentially thinner, a solid containing phosphor, and a tritiated binder. The tritium shall be chemically combined with the binder. The amount of tritium (Hydrogen 3) used to obtain the desired brightness shall not exceed the number of millicuries as required by detailed procedures, drawings and specifications. The luminous material shall be used within sixty days of its formulation. The material used to encapsulate and seal the luminous material shall be transparent and capable of meeting all requirements herein specified.

3.4 Luminous sealed sources.- Luminous sealed sources made with radioactive luminous compounds shall be in accordance with Government approved drawings. The fluorescent color of the luminous compound shall conform to Government approved detailed drawings and specification requirements. When the compound is applied to a surface the initial brightness of the luminous sector shall be not less than the number of effective microlamberts specified on the Government approved drawings.

3.5 Contamination.- All exposed surfaces of the sealed material shall be wiped with high wet strength filter paper. The beta radioactivity accumulated on the filter paper shall at no time exceed 0.05 microcurie.

3.6 Storage temperature.- The sealed tritium activated material shall be exposed and thermally stabilized for a period of two hours at atmospheric pressure at an ambient temperature of minus eighty degrees Fahrenheit, and for a period of two hours at atmospheric pressure at an ambient temperature of plus one hundred sixty degrees Fahrenheit. Subsequent to exposure at these temperatures, the sealed material shall show no evidence of damage in any form to the luminous area and shall be capable of meeting the requirements of 3.5.

3.7 Altitude.- Sealed material which has been subjected to the requirements of 3.6 shall be exposed and thermally stabilized at an ambient temperature of minus sixty five degrees Fahrenheit for a period of four hours at a pressure of eighty three millimeters of mercury, absolute. The temperature shall then be raised over a period of six hours to plus eighty six degrees Fahrenheit for a period of four hours at a pressure of one hundred thirty eight millimeters of mercury, absolute. Subsequent to which the temperature shall be reduced to plus

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

sixty eight degrees Fahrenheit at a pressure of eighty three millimeters of mercury, absolute and maintained for a period of four hours. Subsequent to meeting this requirement the sealed material shall show no evidence of damage in any form to the luminous area and shall be capable of meeting the requirements of 3.5.

3.8 Vibration.- The sealed material shall be subjected to vibration at a constant acceleration of 2.5 gravitational units for three minutes at each of the four most severe resonant frequencies. The four most severe resonant frequencies shall be determined by proper instrumentation during exposure to two vibratory cycles of fifteen minutes duration each over the frequency range of 5 to 500 cycles per second. Subsequent to vibration the requirements of 3.5 shall be met.

3.9 Shock.- The sealed material shall be subjected to three 150 G shocks, in each of three mutually perpendicular directions. Subsequent to the three shocks the sealed material shall show no evidence of damage in any form to the luminous area and shall be capable of meeting the requirements of 3.5.

3.10 Solubility.- Subsequent to being subjected to the requirements of 3.6, 3.7, 3.8 and 3.9, the sealed material containing tritium shall be submerged in water, the ambient temperature of which is between plus 60 to 90 degrees Fahrenheit, for a period of twenty-four hours. Subsequent to the water submersion not more than 0.05 microcurie of the luminous compound shall be dissolved in the water.

3.11 Workmanship.- Workmanship shall be of the highest grade throughout and in accordance with best laboratory and commercial practices for the manufacture, application and sealing of tritium material. All finishes shall be free from flaking or chipping of the luminous material or failure of the overlap seal or other physical defects that would tend to release or disseminate the luminous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

4.2 Sampling.- Sampling shall be in accordance with Standard MIL-STD-105, unless otherwise specified.

4.3 Material.- The luminous tritium activated material shall be inspected by laboratory methods to determine that the processes and homogeniety of the materials and binder and the transparency of the encapsulating material is in compliance with 3.3 and the specified drawings and specifications.

4.4 Identification license and disposition.- Sealed material shall not be submitted for final acceptance inspection without proof of licensing in compliance with 3.2. Items containing tritium activated luminous material shall be visually inspected to ascertain that the specified markings are in the immediate area of the luminous material in compliance with 3.3 and comply with regulations governing the identification and marking of tritium activated material as stated in 3.2.

4.5 Luminous sealed sources.- Sealed material containing radioactive luminous compounds shall be tested with equipment capable of determining the brightness in effective microlamberts as specified in the detailed specification or drawings in compliance with 3.4.

4.6 Contamination test.- Subsequent to each of the tests in 4.7 to 4.10 inclusive, all exposed surfaces of the sealed material shall be thoroughly wiped with a piece of filter paper of high wet strength and absorption capacity, moistened with a solution which will not attack the material of which the sealed source is made and which, under the conditions of this test, has been demonstrated to be effective in removing the radioisotope involved. The paper shall be allowed to dry then the amount of tritium on the paper shall be determined by a testing device that has been calibrated to measure the accumulated beta radioactivity in microcuries, in compliance with 3.5.

4.7 Temperature test.- The sealed material shall be placed in a test chamber and subjected to the extreme temperatures and required time limit specified to determine compliance with 3.6. At the conclusion of this test, it shall be visually inspected for any damage to the luminous area and shall then be subjected to the test of 4.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

4.8 Altitude test.- The sealed material shall be placed in a test chamber and thermally stabilized at the specified temperatures and pressures for the required length of time to determine compliance with 3.7. At the conclusion of this test the sealed material shall be visually inspected for any damage to the luminous area and shall then be subjected to the test of 4.6.

4.9 Vibration.- The sealed material shall be secured on a vibration test stand and vibrated in accordance with the requirements of 3.8. Subsequent to vibration the item shall be subjected to a visual and tactile examination for damage and shall then be subjected to the test of 4.6.

4.10 Shock test.- The sealed material shall be shock tested in accordance with Method 202 of Standard MIL-STD-202. Subsequent to the shock test the sealed material shall be inspected for damage to the luminous area to determine compliance with 3.9, and the item shall then be subjected to the test of 4.6.

4.11 Solubility test.- When testing sealed material to determine solubility in compliance with 3.10 it shall be immersed in water at the specified temperature for the required time. After the water is evaporated, the residue shall then be measured for the microcurie content which shall not exceed the amount specified in 3.10. This shall be the final test performed on the sealed material.

4.12 Workmanship.- All finishes shall be inspected to ascertain that there is freedom from flaking or chipping of the luminous material, or failure of the seal to overlap the material which would tend to release or disseminate the luminous material.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with requirements furnished by the contracting officer in connection with the contract and shall specify details for handling, packaging, packing and storage of material containing tritium (hydrogen 3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

5.2 Identification markings.- Identification markings for shipment and storage of individual items or bulk shipments shall show the following as permanent markings.

(a) Radiation Caution Symbol
(b) Atomic Energy Commission License Number
(c) Contains Mc of Radioactive H^3
(d) If Found Return to Military Authority

6. NOTES

6.1 Intended use.- This general specification is intended to establish the basic safety requirements and tests required of licensed producers of tritium encapsulated sealed self luminous material utilized by agencies within the Department of Defense.

6.2 For informational purposes and guidance, Atomic Energy Commission announcement "RC-13 Sealed Sources Containing Beta and/or Gamma-Emitting Radioisotopes", may be used for reference.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodian:

Army-Ord

Preparing activity:

Army-Ord

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.